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Inria & CETIM Research Team

Subject: Application for PhD Position – Hybridization of simulation and learning models in mechanical systems

Dear Inria and CETIM Research Team,

I am applying for the PhD position on "Hybridization of simulation and learning models in mechanical systems" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

Developing a unifying theory for hybrid modeling is a challenge I am deeply passionate about. My engineering of the Industrial Gas Turbine Digital Twin directly addressed this by combining traditional thermodynamic simulations with Deep Learning anomaly detection. I am eager to bring this practical experience to CETIM and Inria, researching foundational abstractions that allow learning models and physical simulations to interoperate seamlessly.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui